

Experiment 11

K-Nearest Neighbors (KNN).

Program:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Iris.csv')
df.info()
df.variety.value_counts()
df.head()

features=df.iloc[:, :-1].values
label=df.iloc[:, 4].values

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

xtrain,xtest,ytrain,ytest=train_test_split(
features, label, test_size=.2, random_state=1)

model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)

print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))

from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))

from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))
```

Output

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
0   sepal.length    150 non-null    float64
1   sepal.width     150 non-null    float64
2   petal.length    150 non-null    float64
3   petal.width     150 non-null    float64
4   variety         150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
variety
Setosa      50
Versicolor  50
Virginica   50
```

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

Scores:

```
0.9583333333333334
1.0
```

Confusion Matrix:

```
[[50  0  0]
 [ 0 47  3]
 [ 0  2 48]]
```

Classification Report:

	precision	recall	f1-score	support
Setosa	1.00	1.00	1.00	50
Versicolor	0.96	0.94	0.95	50
Virginica	0.94	0.96	0.95	50
accuracy			0.97	150
macro avg	0.97	0.97	0.97	150
weighted avg	0.97	0.97	0.97	150